BIOS vs UEFI Comparison

Feature	BIOS	UEFI
Definition	Traditional firmware interface used to initialize hardware and start the operating system	Modern firmware interface designed to replace traditional BIOS
Boot Process	Performs hardware initialization and searches for bootable devices	Performs hardware initialization and uses boot managers stored in the EFI System Partition
Maximum Disk Support	Traditional BIOS systems commonly use MBR, which has a practical 2 TB disk limitation	UEFI commonly uses GPT, supporting very large disks
Partition Style	Traditionally MBR	Commonly GPT
Security Features	Limited compared with modern firmware	Supports features such as Secure Boot
Speed	Generally slower during boot	Generally faster and more efficient on modern systems
Advantages	Simple, widely supported by older hardware, familiar technology	Modern architecture, GPT support, Secure Boot, better support for large drives
Disadvantages	Older architecture, MBR limitations, fewer modern security features	More complex and may have compatibility issues with very old operating systems
Modern Usage	Mostly found on legacy systems	Standard firmware technology on most modern computers

Why UEFI Has Replaced BIOS

UEFI has largely replaced traditional BIOS because modern computers require capabilities that the older BIOS architecture was not designed to provide. Traditional BIOS was developed for an earlier generation of computers and commonly works with the MBR partitioning scheme. MBR has limitations when dealing with large storage devices and provides fewer capabilities for modern system management.

UEFI provides a more modern firmware environment. One of its important advantages is its ability to work with GPT partitioning, which supports much larger storage devices and more partitions than traditional MBR. This makes UEFI more appropriate for modern computers with large storage capacities.

UEFI also provides improved security capabilities. Secure Boot can help ensure that trusted boot software is loaded during startup, reducing the risk of unauthorized boot-level software interfering with the operating system.

Another advantage is that UEFI provides a more flexible boot environment. Instead of relying solely on the traditional BIOS boot process, UEFI can interact with boot managers stored on the EFI System Partition. This architecture makes modern operating system boot management more flexible.

For these reasons, UEFI is now the standard firmware technology used by most modern computers. BIOS remains important for understanding legacy systems, troubleshooting older computers, and supporting older hardware, but UEFI is generally the preferred technology for new systems.